

→ SLR(1) Parsing :-

SLR stands for :-

Simple LR

and (1) means it uses one-symbol lookahead indirectly through FOLLOW sets.

So ;

SLR(1) = Simple LR with 1 lookahead.

SLR uses the FOLLOW set to decide where a completed LR(0) item should reduce.

→ Why do we need SLR(1) ?

LR(0) problem.

Suppose we have :-

$C \rightarrow d$.

LR(0) says :

R3

for every terminal :-

State	c	d	\$
I4	R3	R3	R3

• why ?

~~Because~~
Because LR(0) has zero lookahead.

SLR(1) improve this by asking:

⇒ "After c, which symbols are actually allowed to appear?"

We find

FOLLOW(c)

Eg;

let's take the same grammar.

$S \rightarrow CC$

$C \rightarrow cC$

$C \rightarrow d$

⇒

Augmented grammar:

$S' \rightarrow S$

$S \rightarrow CC$

$C \rightarrow cC$

$C \rightarrow d$

We already constructed the LR(0) states and GOTO transitions.

All things are same as we done before in LR(0). In this we have to change only those important states { Reduce entries { $\cdot$ at the end } }.

Because the reduction is placed under all terminal columns.

from our given canonical collection:

Important states are:

$I_4 : C \rightarrow \cdot$

$I_5 : S \rightarrow \cdot CC$

$I_6 : C \rightarrow c \cdot C$

So we have to find **Follow Sets**:

So we have to find follow of S and c
Grammar:

$S \rightarrow CC$

$C \rightarrow CC$

$C \rightarrow c$

Now,

$$\underline{\text{FOLLOW}(S) = \{\$ \}}$$

$$\text{FOLLOW}(C) = ?$$

$$\text{Now } S \rightarrow C C$$

$$\begin{array}{c} \uparrow \quad \quad \quad \nearrow \\ \text{FOLLOW}(C) = \text{FIRST}(C) \end{array}$$

$$\text{FIRST}(C) = \{c, d\}$$

~~FOLLOW(C) = FOLLOW(S)~~

$$S \rightarrow C C$$

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$$\text{FOLLOW}(C) = \text{FOLLOW}(S)$$

$$\therefore \underline{\text{FOLLOW}(C) = \{c, d, \$ \}}$$

Now,

Construct SLR table.

The shift and Goto entries remain the same as LR(0).

The important change is **Reduce**.

→ I_4 :

$C \rightarrow d$.

Reduce by $C \rightarrow d$ {production no. 3}

So $FOLLOW(C) = \{ c, d, \$ \}$

State	c	d	\$
I_4	R3	R3	R3

→ I_5 :

$S \rightarrow CC$.

Reduced by $S \rightarrow CC$ {production 1}

$FOLLOW(S) = \{ \$ \}$

State	c	d	\$
I_5			R1

→ I_6 :

$C \rightarrow cC$.

Reduced by $C \rightarrow cC$ {production 2}

So

$$\text{Follow}(e) = \{ c, d, \$ \}$$

State	c	d	\$
T6	R2	R2	R2

• SLR(1) Table :-

State	c	d	\$	S	C
0	S3	S4		1	2
1			ACCEPT		
2	S3	S4			5
3	S3	S4			6
4	R3	R3	R3		
5			R1		
6	R2	R2	R2		

→ This become useful when there is conflict in LR(0) in our eg there is no conflict. Suppose from our table [4, d] has both reduce and shift or ~~reduce/reduce~~ reduce/reduce in LR(0) then SLR(1) resolves that problem.

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